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Bihar Engineering University, Patna

B.Tech 1st Semester Exam-2022

Time: 03 Hours

Code: 102101 ***Subject: Physics (Electromagnetism)*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.
(v) Symbols used (if any) have their usual meanings.

Q.1 Answer any seven of the following: **[2x7=14]**

(a) Define electric polarization.

(b) Write down Laplace's equation.

(c) Define displacement current.

(d) What is the physical interpretation of bound charges?

(e) Define diamagnetism. Give two examples of diamagnetic materials.

(f) With necessary expression, explain standing wave ratio.

(g) What do you mean by skin effect?

(h) Explain the terms motional e.m.f. and transformer e.m.f.

(i) Differentiate between conduction and convection current.

(j) What is meant by retarded potential?

Q.2 Solve both questions : **[7+7=14]**

(a) Find the electric field at a distance z above the centre of a circular loop (radius R) carrying uniform linear charge λ.

(b) Write down the expression for electric field due to surface charge distribution of volume charge density ρ.

Q.3 Solve this question : **[14]**

Derive the expression for Transmission coefficient of electromagnetic waves from a non-conducting medium–vacuum interface for normal incidence.

Q.4 Solve both questions : **[7+7=14]**

(a) A point charge q is situated at a distance a from the centre of a grounded conduction sphere of radius R. Using the method of images, find the potential outside the sphere.

(b) Explain Faraday cage? What is the electrical force inside a Faraday cage when it is struck by lightning?

Q.5 Solve both questions : **[7+7=14]**

(a) Derive continuity equation for current densities.

(b) State and derive Poynting theorem.

Q.6 Solve both questions : **[7+7=14]**

(a) Derive the boundary conditions for electrostatic field intensity and electric flux density at (i) the interface between two dielectrics and (ii) the interface between a perfect conductor and a dielectric.

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(b) A long spherical cloud of radius r has a uniform volume charge distribution of ρ_v. Calculate the potential distribution and the electric field at any point in space using Poisson's and Laplace's equations.

Q.7 Solve both questions : **[7+7=14]**

(a) A solenoid of radius 4 mm and length 2 cm has 150 turns/m and carries current 500 mA. Find- (i) |H| at the centre (ii) |H| at the ends of the solenoid.

(b) Determine whether the following potential equations satisfy Laplace's equation or not: (i) $V = 2x^2 - 4y^2 + z^2$ (ii) $V = r^2\cos\varphi + \theta$.

Q.8 Solve both questions : **[7+7=14]**

(a) State Ampere's circuit law. Write its application.

(b) A hollow conducting cylinder has inner radius a and outer radius b and carries current I along the positive z-direction. Find H everywhere.

Q.9 Write short notes on any two of the following: **[7x2=14]**

(a) Transverse nature of electromagnetic waves propagating in vacuum
(b) Energy carried by electromagnetic waves
(c) Electromagnetic braking and its application
(d) Electric field due to electric dipole

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